

SHUBHAM DINESH BAFNA

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EDUCATION

University of Southern California

Master of Science in Computer Science – GPA: 3.33/4.0

Los Angeles, USA

January 2023 - December 2024

Courses: Analysis of Algorithms, Machine Learning, Web Technologies, Foundations of AI, Applied NLP, Database Systems, Information Retrieval

University of Mumbai

Bachelor of Engineering in Computer Engineering – GPA: 9.33/10

Mumbai, India

August 2018 - July 2022

TECHNICAL SKILLS

- **Programming Languages:** Swift, Python, Java, C, C++, JavaScript, TypeScript, PHP, SQL
- **Web/App Development:** SwiftUI, watchOS, CoreMotion, HTML5, CSS3, Bootstrap, Angular, React, Node.js, Flask, Android SDK, Redux, WebSocket
- **Databases & Cloud Platforms:** MySQL, MongoDB (including Mongoose), PostgreSQL, CloudKit, AWS (EC2, S3, Lambda), GCP (Compute, Storage)
- **Tools:** Xcode, Xcode Instruments, Git/GitHub, Android Studio, Docker, Jenkins, Postman, Tableau, Jira, Trello, Slack
- **Skills:** Full-Stack Web Development, Object-Oriented Programming, Multithreading, Agile, Project Management, A/B Testing, ML, NLP

WORK EXPERIENCE

University of Southern California

Los Angeles, USA (Hybrid)

Software Engineer (Gen-AI)

February 2025 – Present

- Engineered a text-based AI voice synthesizer in React, Python, and Flask, leveraging ElevenLabs API for high-fidelity real-time voice generation, improving voice match accuracy and maintainability.

Stealth Startup

Los Angeles, USA

iOS Developer Intern

September 2024 – January 2025

- Converted an existing website into an iOS app, designing the interface and implementing the frontend using Swift, reducing development time by 40% through efficient UI/UX planning and GCP integration.

WorkUp

Los Angeles, USA

Software Engineering Intern

May 2024 - August 2024

- Migrated the platform from HTTP to WebSocket using React, Node, & Redux on GCP, reducing latency by 78% and enhancing audio/video clarity for real-time calls
- Researched and replaced Azure Communication Services SDK with Zoom SDK to achieve 60% financial savings
- Spearheaded a 5-member team, enhancing development efficiency by 20% through effective task management on Trello
- Managed weekly code merges and expedited critical backend issue resolution between AI and iOS teams using Node & GCP, reducing downtime by 15%
- Built features from scratch using JavaScript to eliminate paid SDK dependencies, enhancing system security and maintainability

River Route Creative Group LLP

Mumbai, India

Software Engineer

November 2021 - December 2022

- Implemented a full-stack lung cancer doctor-patient discussion assistant tool for Roche using React, Node.js, Express, AWS, MySQL, MongoDB, Bootstrap, and dynamic content integration of clinical data, resulting in a significant 51% increase in patient retention for Roche
- Integrated widgets like Magnific Popup and Owl Carousel, while developing a REST API to collect and manage real-time data regarding patients' estimated life expectancy
- Created a database for gathering research responses from doctors, enabling a 33% reduction in the research approval time

PROJECTS

Maby – Baby Tracker | Swift, SwiftUI, CoreMotion, CloudKit

January 2025

- Developed a baby tracker app for iOS and watchOS using Swift & SwiftUI with a modular UI, reducing data logging time by 36%
- Engineered efficient sensor polling with CoreMotion, cutting CPU usage by 21% and extending battery life
- Integrated CloudKit for cross-device sync with robust error handling, achieving 96% real-time sync success rate
- Profiled performance via Xcode Instruments, improving UI response speed by 28%

Stock Search Web Application | Angular, Android (Java), iOS (SwiftUI), Node.js, MongoDB Atlas, Flask

May 2024

- Created a single-page responsive Angular web application and native Android app with real-time stock data updates every 15 seconds throughout the website via the Finnhub API and Polygon.io API, achieving 33% faster data retrieval and 35% improved user interaction through dynamic data visualization with Highcharts
- Incorporated lazy loading of 10 results at a time, ensuring a seamless user experience, worked on REST APIs using Node.js, and deployed on GCP
- Extended the platform to iOS with SwiftUI 5 (MVVM), achieving 40% code reuse compared to building a standalone Swift project
- Employed WKWebView for interactive charts, integrating the existing Node.js + MongoDB for response times under 250ms and real-time data consistency across Web, Android, and iOS

Pente AI MiniMax Game Agent | Game Agent, Decision Making, Optimization

April 2023

- Engineered a Pente game agent using the Minimax algorithm with Alpha-Beta pruning, achieved a 97% win rate against [Pente.org's](#) AI bot, and consistently outperformed human players
- Leveraged a hash table instead of standard NumPy arrays, enhancing the bot's search speed by 9 times to conduct a 4-depth search within 10 seconds across evolving game grid spaces ranging from 7x7 to 19x19 in Pente